DERIVED CHOW'S LEMMA FOR A SEPARATED ALGEBRAIC SPACE OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12033-12112, immediately following the statement at 12012-12031.

Editorial note. Same-name pair note: this is the more general (algebraic spaces) side; the schemes side is not counted beside it.

Editorial note. Six cross-chapter pointers (Sites and Sheaves, Limits of Spaces, Morphisms of Spaces, Derived Categories of Spaces) plus one same-chapter lemma are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Descending finite presentation and separatedness along a limit. Limits of Spaces, lemmas spaces-limits-lemma-descend-finite-presentation and spaces-limits-lemma-descend-separated-morphism, cited in the proof. Moves the situation to a Noetherian approximation where the construction can be done.
- Scheme-theoretic image of a partial section. Morphisms of Spaces, lemma spaces-morphisms-lemma-scheme-theoretic-image-of-partial-section, cited in the proof. Produces the closed subspace Z inside X x_A P^n_A with the required property over the open V.
- Identifying a pushforward as pseudo-coherent in the Noetherian case. Derived Categories of Spaces, lemmas spaces-perfect-lemma-compare-base-change and spaces-perfect-lemma-identify-pseudo-coherent-noetherian, cited in the proof. Supplies the object E and its pseudo-coherence, which is what the derived statement adds to Chow's lemma.
- Pseudo-coherence under pullback. Sites and Sheaves, lemma sites-cohomology-lemma-pseudo-coherent-pullback, cited in the proof. Keeps the pseudo-coherence conclusion stable along the base changes used in the construction.
- Tor independence in the base change square. chapter lemma lemma-case-of-tor-independence, cited in the proof. Controls the derived tensor products appearing in the construction of b and c.

Source reference. Original lemma 0CTQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11969–11969. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08H4: focused-seeds/geometry/S9-spaces-derived-chow-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 07SK: focused-seeds/geometry/S9-spaces-derived-chow-spaces/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0851: focused-seeds/geometry/S9-spaces-derived-chow-spaces/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CP3: focused-seeds/geometry/S9-spaces-derived-chow-spaces/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08IR: focused-seeds/geometry/S9-spaces-derived-chow-spaces/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08IK: focused-seeds/geometry/S9-spaces-derived-chow-spaces/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (Derived Chow's lemma)**.** *Let A be a ring. Let X be a separated algebraic space of finite presentation over A. Let $x \in |X|$. Then there exist an $n \geq 0$, a closed subspace $Z \subset X \times_A \mathbf{P}^n_A$, a point $z \in |Z|$, an open $V \subset \mathbf{P}^n_A$, and an object E in $D(\mathcal{O}_{X \times_A \mathbf{P}^n_A})$ such that*

(1) *$Z \to X \times_A \mathbf{P}^n_A$ is of finite presentation,*
(2) *$c : Z \to \mathbf{P}^n_A$ is a closed immersion over V, set $W = c^{-1}(V)$,*
(3) *the restriction of $b : Z \to X$ to W is étale, $z \in W$, and $b(z) = x$,*
(4) *$E|_{X \times_A V} \cong (b, c)_* \mathcal{O}_Z|_{X \times_A V}$,*
(5) *E is pseudo-coherent and supported on Z.*

Proof. We can find a finite type $\mathbf{Z}$-subalgebra $A' \subset A$ and an algebraic space X' separated and of finite presentation over A' whose base change to A is X. See Limits of Spaces, Lemmas 07SK and 0851. Let $x' \in |X'|$ be the image of x. If we can prove the lemma for $(X'/A', x')$, then the lemma follows for $(X/A, x)$. Namely, if n', Z', z', V', E' provide the solution for $(X'/A', x')$, then we can let $n = n'$, let $Z \subset X \times \mathbf{P}^n$ be the inverse image of Z', let $z \in Z$ be the unique point mapping to x, let $V \subset \mathbf{P}^n_A$ be the inverse image of V', and let E be the derived pullback of E'. Observe that E is pseudo-coherent by Cohomology on Sites, Lemma 08H4. It only remains to check (5). To see this set $W = c^{-1}(V)$ and $W' = (c')^{-1}(V')$ and consider the cartesian square

$$\begin{array}{ccc} W & \longrightarrow & W' \\ {\scriptstyle (b,c)}\downarrow & & \downarrow{\scriptstyle (b',c')} \\ X \times_A V & \longrightarrow & X' \times_{A'} V' \end{array}$$

By Lemma 0CTQ $X \times_A V$ and W' are tor-independent over $X' \times_{A'} V'$. Thus the derived pullback of $(b', c')_*\mathcal{O}_{W'}$ to $X \times_A V$ is $(b, c)_*\mathcal{O}_W$ by Derived Categories of Spaces, Lemma 08IR. This also uses that $R(b', c')_*\mathcal{O}_{Z'} = (b', c')_*\mathcal{O}_{Z'}$ because (b', c') is a closed immersion and similarly for $(b, c)_*\mathcal{O}_Z$. Since $E'|_{U' \times_{A'} V'} = (b', c')_*\mathcal{O}_{W'}$ we obtain $E|_{U \times_A V} = (b, c)_*\mathcal{O}_W$ and (5) holds. This reduces us to the situation described in the next paragraph.

Assume A is of finite type over $\mathbf{Z}$. Choose an étale morphism $U \to X$ where U is an affine scheme and a point $u \in U$ mapping to x. Then U is of finite type over A. Choose a closed immersion $U \to \mathbf{A}^n_A$ and denote $j : U \to \mathbf{P}^n_A$ the immersion we get by composing with the open immersion $\mathbf{A}^n_A \to \mathbf{P}^n_A$. Let Z be the scheme theoretic closure of

$$(\mathrm{id}_U, j) : U \longrightarrow X \times_A \mathbf{P}^n_A$$

Let $z \in Z$ be the image of u. Let $Y \subset \mathbf{P}^n_A$ be the scheme theoretic closure of j. Then it is clear that $Z \subset X \times_A Y$ is the scheme theoretic closure of $(\mathrm{id}_U, j) : U \to X \times_A Y$. As X is separated, the morphism $X \times_A Y \to Y$ is separated as well. Hence we see that $Z \to Y$ is an isomorphism over the open subscheme $j(U) \subset Y$ by Morphisms of Spaces, Lemma 0CP3. Choose $V \subset \mathbf{P}^n_A$ open with $V \cap Y = j(U)$. Then we see that (2) holds, that $W = (\mathrm{id}_U, j)(U)$, and hence that (3) holds. Part (1) holds because A is Noetherian.

Because A is Noetherian we see that X and $X \times_A \mathbf{P}^n_A$ are Noetherian algebraic spaces. Hence we can take $E = (b, c)_*\mathcal{O}_Z$ in this case: (4) is clear and for (5) see Derived Categories of Spaces, Lemma 08IK. This finishes the proof. □